Reading Time: An initial **2 minutes**

MATHEMATICS METHODS : UNITS 1 & 2, 2020

Test 1 – (10%)

1.2.1 – 1.2.8

Time Allowed 45 minutes	**First Name**	**Surname**	**Marks** 43 marks

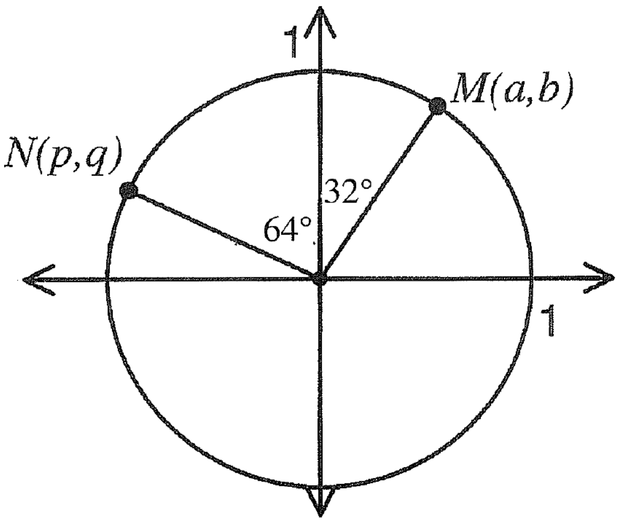
Circle your Teacher's Name: Benko Bestall Fraser-Jones Goh Koulianos Luzuk Rudland Tanday

Assessment Conditions: ***(N.B. Sufficient working out including diagrams must be show to gain full marks)***

- Calculators: Allowed
- Formula Sheet: Provided
- Notes: Not Allowed

Question 1

(1, 1, 1 & 1 = 4 marks)



Two points *N(p,q)* and *M(a,b)* are plotted on the unit circle above.
In terms of *a*, *b*, *p and/or q,* determine the following.

i) $\sin 26^\circ$

ii) $\cos 154^\circ$

iii) $\cos 26^\circ$

iv) $\tan 122^\circ$

Question 2

(2, 2 = 4 marks)

(a) Express the following angles in radians. Answers should be simplified, in exact values and in terms of π .

(i) 45°

(ii) 160°

(b) Express the following angles in degrees.

(i) $\frac{3\pi}{5}$ radians

(ii) 2.2 radians (round to nearest degree)

Question 3

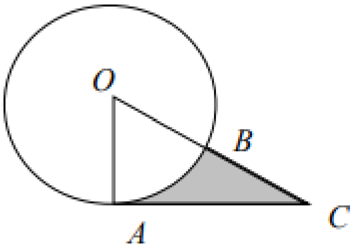
(5 marks)

A parallelogram has side lengths of 25 and 30 cm and an area of 700cm^2. Find the length of the longest diagonal.

Question 4

(4 marks)

Triangle *ACO* is a right angled triangle with *OA* = 10cm, *AC* = 20cm and the size of angle *OAC* = 90˚. A circle of radius 10cm is drawn, centre *O*. Find the area of that part of the triangle *OAC* not lying in the circle.



Question 5

(4 marks)

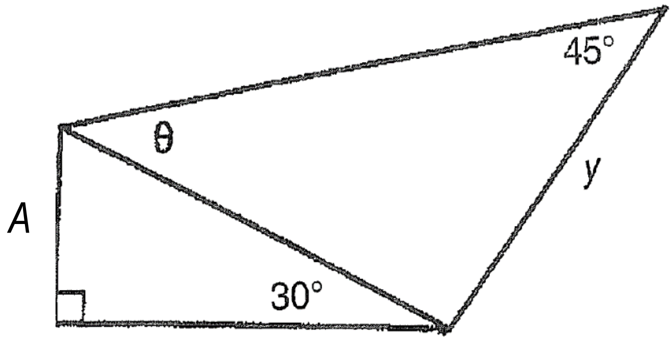
Points *A* and *B* lie on the circumference of a circle with centre *O* and radius 12cm. If the major arc *AB* has a length of 60cm, find the area of the minor sector *AOB*.

Question 6

(4 marks)

For the diagram on the right show that

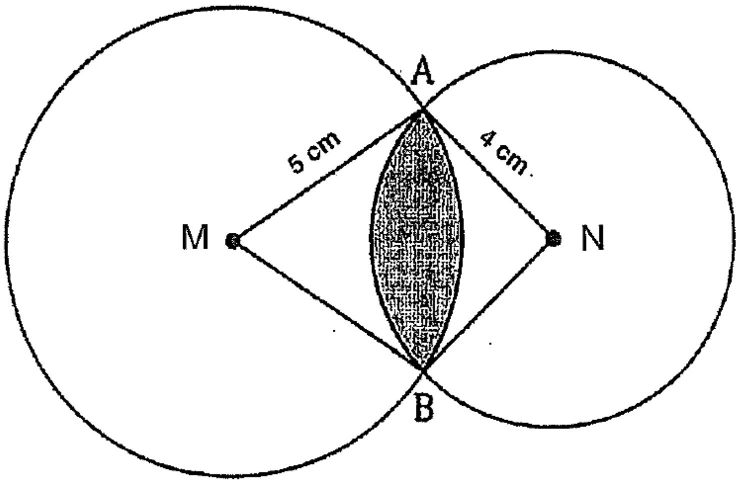
$$y = 2\sqrt{2}\ A\sin\theta$$



Question 7

(6 marks)

The diagram below shows two overlapping circles of radii 4 cm and 5 cm, where the size of angle *MAN is* 80°. Determine the area of the region common to both circles.



Question 8

(6 marks)

Ken and Sofia set out in different directions from their campsite. Ken walks 2150m on a bearing of 316°, whilst Sofia walks for 2000m on an unknown bearing. When they finish walking the distance between them is 2696m. What are the possible bearings that Sofia could have used?

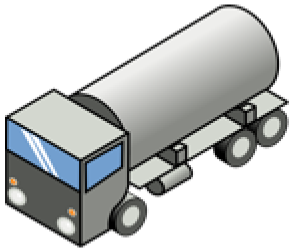
Question 9

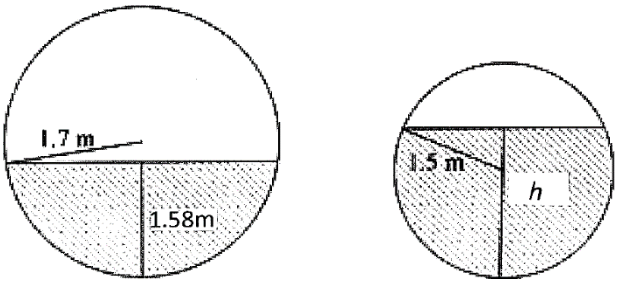
(6 marks)

A tanker, truck A, is loaded with petrol however receives a tyre puncture. Another tanker that is empty, truck B, is sent to be loaded with all the petrol from the tank of truck A. Both trucks have cylindrical tanks.

Truck A has a tank of radius 1.7 m and a length of 6 m. The tank is filled with petrol to a height of 1.58 m.

Truck B has the same length but a radius of 1.5m.





Circular ends of the petrol tanks.

The drivers are concerned as Truck B has a smaller tank. How high (h) will the petrol reach in the tank of Truck B?

End of Paper